

cost effective techniques

- cost effective housing techniques are mainly focus on reduce the construction costs by using advanced and new building materials.
- quality & safety [by maintaining these we can reduce the construction costs]

Housing scenario :-

- housing market trends,
 - housing affordability,
 - homeless.
 - housing policies.
- If we construct the houses by using the eco-friendly materials like bamboo, or recycled materials.

- Another option is modular construction.

↓ construction.
reduce costs &
construction time.

- India's housing scenarios are mainly characterised by the large population,

Here are some factors that contribute the housing shortage :-

factors

1] population growth

2] urbanisation

3] migration from rural areas.

1] availability
2] affordability
3] economic factors.
4] quality
5] social

Status of urban housing

- urban housing typically have some challenges such as

* high population density

* limited space.

* lack of land and housing.

* high construction costs

* inadequate plan and infrastructure.

} shortage of affordable housing

Status of rural housing :-

→ In rural areas, houses can be constructed with occupying more space.

→ There is a more gap b/w the one house to another house.

→ Rural houses means houses are constructing in rural areas like villages, hilly areas.

→ Challenges :-

* Insufficient funding

* Inefficient transportation

* Lack of higher education.

* ^{lack of} medical facilities

* Lack of basic things like

1) clean water

2) Electricity

3) sanitation.

* Infrastructure

* ^{limited} financial resources [hard to use quality housing materials & construction techniques]

* Climate & Natural disasters.

↳ heavy rainfall

↳ earthquake
↳ hurricanes
↳ floods.

* finding skilled labours.

* cultural and social considerations.

→ 50 % of the houses in India are good in condition.

* 1] Pradhan Mantri Awas Yojana

2] Indira Awas Yojana.

→ In India, Agriculture is the primary industry, most of the people in India are live on their farms.

Housing Finance :-

→ housing finance is a loans which are helped to the people to buy the residential properties like houses, buildings, lands etc.

→ Also improved already existing properties.

→ Floccing loans can be given by the banks
some of top home loan banks in India.

- 1] HDFC home loan
- 2] AXIS Home loan
- 3] ICICI " "
- 4] Bank of baroda
- 5] citibank home loan.

→ HFC's housing finance companies can provide maximum of 1 crore to each individual to purchase (or) repair.

key points: Apartment, homes, buildings.

- home loans

- 2] down payment (most home buyers are forced to use this method) some money

→ Initial down payment (first paid)
 & Remaining amount can pay
 rates ÷ through housing finance.

- 3] Interest rates $\frac{\circ}{\circ}$

- i) fixed rate mortgages [stable]

- ii] Variable rate mortgages [changes over time]

4) Repayment terms :

→ The period of start ^{of} credit to the end of the transaction.

→ housing loans have various repayment terms such as
15 years
20 years
30 years.

→ There are two choices to pay the interest
* monthly interest

* total interest paid at end

5) credit worthiness : ^{of the transaction.} Payoff of money.

→ lender's asses the credit worthiness of borrowers before approving a housing loan
It depends on 5 factors

a) character [borrower keeps promises]

b) capital [borrower has any lands]

c) capacity [how much money borrower can handle in the past].

d) conditions

e) collateral

6) Government programs :

→ many government can offer HF programs

such as i) provide subsidies

ii) low-interest loans

iii) Affordable housing

7) Real estate market

8) Refinancing [again taking the loans]

9) Professional advices

2] Existing finance system in India :

→ diagram

- 1) Banking
- 2) Non-banking finance systems
- 3) stock markets
- 4) mutual funds
- 5) capital markets
- 6) payment systems
- 7) Regulatory bodies
- 8) Government initiatives
- 9) Insurance sector
- 10) microfinance.

3] Government as a role facilitator

subsidized

non-subsidized

- Available for undergraduates
- lower loan limits
- Government pay interest when students enrolled

- Available for undergraduates & graduates.
- higher loan limits
- Govt can not help in interest

- i] subsidized interest rates
- ii] public-private partnership
- iii] loan guarantees
- iv] Affordable housing programs
- v] Regulatory framework [clear & fair]
- vi] Rural & urban housing
- vii] tax incentives
- viii] Financial literacy and education.
- ix] Risk mitigation
- x] credit reporting system.

Status at Rural housing finance

- Government initiatives.
- Interest subsidies.
- financial inclusions.
- housing development programs.
- local conditions. [infrastructure]
- challenges.
- microfinance institutions.

Impediments in housing finance & related issues.

- 1) Infrastructure gaps
- 2) Interest Rates
- 3) Lack of Affordable housing supply
- 4) High Property prices
- 5) Income disparities
- 6) Financial literacy
- 7) environmental sustainability
- 8) Natural disasters.
- 9) Informal housing.
- 10) Government policies.
- 11) Limited access to credit.
- 12) urbanisation challenges.

Land use & Physical planning for housing

- It is an important aspect for urban development and sustainability.
- It is the allocation & organization of land for population to meet the housing needs by considering the environmental, social & Economic factors.

→ This play a crucial role for shaping of land.

main reasons for limited use of land and physical planning for housing.

- 1] Increasing population
- 2] Increasing pollution
- 3] High demand for houses
- 4] Increasing land cost
- 5] Public health issues
- 6] unplanned housing
- 7] slums
- 8] sanitation problems. etc.

planning of urban land

→ In urban areas, planning of land can be done in effective and ethical way.

→ Urban planning very complex to meet the requirements of people.

→ urban planning is the process of designing & developing the urban areas.

key points :-

1] land use zoning :-

urban areas mainly categorizing the land for the purpose of commercial, residential, industrial, and green space.

2] master plans :-

for future development.

3] infrastructure planning

water supply, sanitation.

4] Affordable housing → In urban areas mainly face housing shortages.
→ It can be provided for all income groups.

5] Transportation ÷

6] Environmental sustainability

→ urban planning must consider the environmental factors like green space, air, water quality and climatic conditions.

7] community engagement ÷ good area, local area.

8] land use regulations ÷

→ must consider the factors like size, density, shape of land based on Regulations.

9] safety & security.

→ free from natural disasters.

10] economic development.

→ in future.

Urban land ceiling & regulation act ÷

→ land ceiling is the process of holding the maximum land to each individual.

→ In India, urban land ceiling & regulation act can be proposed in 1976.

→ These act can pass to restrict the amount of vacant land, to each individual, family or private companies.

→ The main objective of this ~~land~~ act is to distribute the land to the poor.

* under this act, urban areas can be divided into 4 classes.

→ In class 'A' towns, a person could own 500 sq meters of land without coming under the ceiling.

→ In 'B' class towns, a person could own 1000 sq meters.

→ In 'C' class 1500 sq mtrs

→ In 'D' class 2000 sq mtrs.

some key aspects under [ULCRA]

1] Identification of urban land :-

Government authorities can mainly identify the urban lands

2] ceiling limit :-

3] Declaration & surrender :-

→ person who have the land excess (or) above ceiling limit, declare to the government (and) pay some fee.

4] Exemptions :- educational, religious, charitable, trusts, old age homes, purposes except ceiling rules.

5] utilization of surplus land :-

→ The surplus land means the land which is received by the land owners. These surplus land can be utilised for excess

many aspects :-

i] distribute^{to} the poor people.

ii] charitable trusts

iii] educational institutions

6] Penalties :-

The person who have the land above ceiling limit [if those lands are identified by the Government authorities, they throw a penalty to those land owners]

7] Zonal Regulations :-

→ These Act can divide the urban areas into 4 towns.

class - A town	→ 500 sqmtrs
class - B town	→ 1000
C	→ 1500
D	→ 2000

→ In 2007, the central government has decided to ULCRA; → distribute to their own states whether they can continue this Act or not.

→ many states can continue this ULCRA for urban development.

→ Now - All the states can follow this ULCRA, but specific regulations & policies may different from one state to another state.

Efficiency of Building bye laws :-

→ ~~Efficiency of~~

Building bye laws is also known as Building Regulations or Building codes.

→ Building bye laws is a set of rules & regulations that govern construction.

→ The main objectives of building bye laws :-
* Building bye laws are important parts of town planning.

* Pre-plan building activities.

* Provide guidelines for development.

- engineers or architects, owners, building

* ensure safety against fire, noise,
structural failures [pollutions].

* The building bye-laws prevent the unplanned development.

Principles and aspects covered in building bye-laws

1] Height of the building.

2] coverage of the building.

3] Green & open spaces [basements]

4] swage facilities [water & power-supply provision]

5] waste management provisions.

[provisions for rainwater harvesting]

6] safety [Building bye laws can priority < fire safety & pollution]

7] adaptability → buildings have chance to change the construction practice

8] Energy efficiency [use Eco-friendly materials → Renewable energy resources]

9] cost effectiveness [safety & less cost]

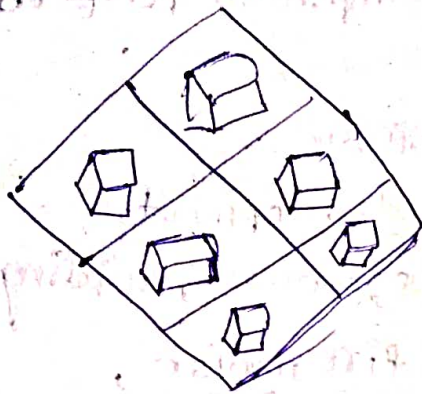
↳ Professional guiding

10] consistency [

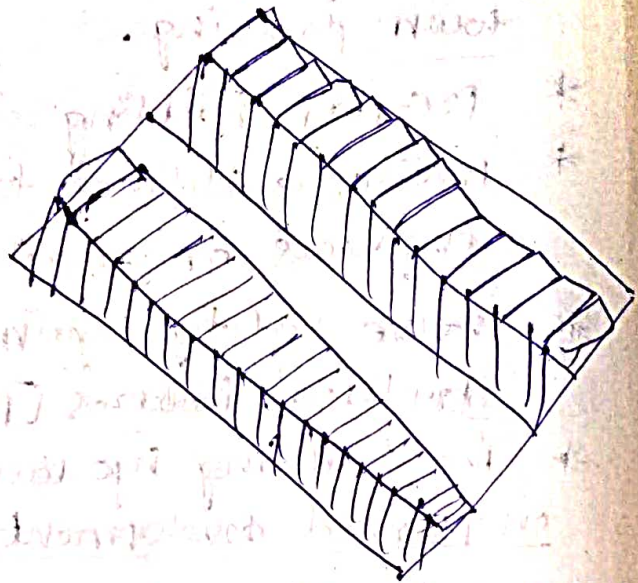
11] public welfare [sanitation, cleaning]

12] environmental impact [reducing waste]

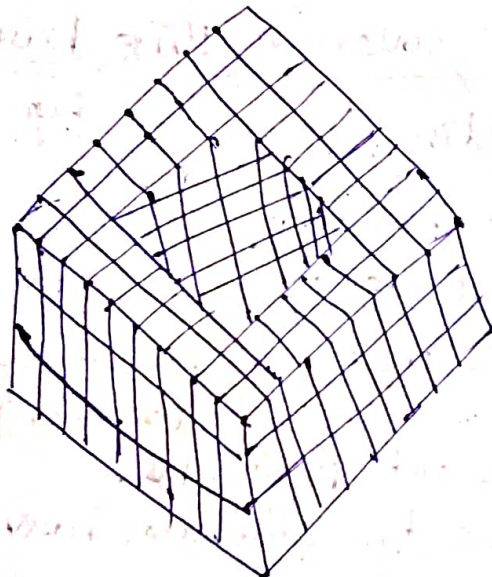
Residential densities :-



Single family
Homes.



Townhomes.



Apartments.

- Residential density is a measure of the no. of people or households in the given particular area.
- It is also expressed as no. of dwelling units per unit area of land.
- Ex :- Residential density can be measured as dwelling units per acre (du/a)

- Residential density is very important for urban planning, because it determines how the density is populated in particular area!
- Residential density is mainly depends on zones. $\left[4 - \frac{P}{S} \right]$

Housing the urban poor :

Living conditions in slums :

- slums means the areas which are far away from the cities or towns.
- Living conditions in slums are widely depends on the factors like
- i] Government policies
 - ii] Location
 - iii] socioeconomic status of the residents.
- slum areas have very dirty and unhygienic surroundings.

causes :

- 1] Over population $\left\{ \begin{array}{l} \text{High population density} \\ \text{multiple families living in small area.} \\ \text{Easy to spread the diseases.} \end{array} \right.$
- 2] migration population [from rural to urban]
- 3] Poverty,
- 4] Natural disaster. [floods, earthquakes etc]
- 5] Inadequate Housing. [houses can be constructed with low quality materials, lack of proper foundation, lack of insulation, ventilation]
- 6] lack of clean water & sanitation
- 7] poor infrastructure [no proper roads, no street lighting]
- 8] Health challenges [

slum area people can face many health problem like Diarrhoea, TB, fever etc (due to lack of sanitation)

- 9] Lack of Transportation facility
- 10] Limited education and opportunity
- 11] Inequality.

Approaches and strategies for housing urban poor.

→ Housing urban poor is a global challenge.

* Here some of the approaches and strategies for housing the urban poor:

- 1] Affordable housing
- 2] slum upgrading [provide clean water & sanitation to the existing slum areas]
- 3] self-help housing [provide land, material & finance]
- 4] Public-private Partnerships [Public developer & private]
- 5] social housing
- 6] micro finance for housing. [small loans & low interest] reduce cost
- 7] land tenure security. [right of people to own the land against Govt evictions]
- 8] Housing subsidies [low-income families]
- 9] Data collection & monitoring.
- 10] Identifying homeless people
- 11] Public Awareness. (PA)
- 12] capacity building (CB)
- 13] Rental housing Regulations.

strategies :-

- 1] Government of India has decided to start the scheme "Housing for All" in the year of 2022.
- 2] 20 million houses will be constructed
- 3] The main objective of this scheme is to avoid homeless people, slum Redevelopment
- 4] Pradhan mantri Awas Yojana
- 5] Indira Awas Yojana.